Unit I: Understanding the data and Installation of R

<table>
<tr><th>Sr. No.</th><th>Question</th><th>Difficulty level</th></tr>
<tr><td>1.</td><td>What is data? What is the difference between data and information? Explain importance of data analysis.</td><td>Medium</td></tr>
<tr><td>2.</td><td>What is data analysis? Explain steps of data analysis.</td><td>Medium</td></tr>
<tr><td>3.</td><td>What are the methods of data collection? Describe in details.</td><td>Medium</td></tr>
<tr><td>4.</td><td>What is the difference between quantitative and qualitative data? Explain their subtypes.</td><td>Medium</td></tr>
<tr><td>5.</td><td>What are advantages of R programming amongst all.</td><td>Medium</td></tr>
<tr><td>6.</td><td>What are applications of Data Analysis? Explain in details.</td><td>Medium</td></tr>
<tr><td>7.</td><td>Give introduction of R software and describe GUI of R-studio</td><td>Medium</td></tr>
<tr><td>8.</td><td>What is difference between R software and R-Studio</td><td>Medium</td></tr>
<tr><td>9.</td><td>Explain all four windows of R-studio along with their functioning</td><td>Medium</td></tr>
<tr><td>10.</td><td>Write in detail about ‘Environment Window’. In dataset what means by observation and variable.</td><td>Medium</td></tr>
<tr><td>11.</td><td>Create a data frame as below
<table>
<tr><th>Name .student</th><th>Percentage</th><th>Class</th></tr>
<tr><td>Namita</td><td>65</td><td>First</td></tr>
<tr><td>Aaroh</td><td>90</td><td>Distinction</td></tr>
<tr><td>Rohit</td><td>46</td><td>Pass</td></tr>
</table></td><td>Medium</td></tr>
<tr><td>12.</td><td>How can you load a .csv file in R? Write syntax of the same.</td><td>Easy</td></tr>
<tr><td>13.</td><td>Write syntax to check inbuilt dataset in R; also details of any dataset out of it.</td><td>Easy</td></tr>
<tr><td>14.</td><td>How do you install and load a package in R?</td><td>Easy</td></tr>
<tr><td>15.</td><td>How can you load a .xls/.xlsx file in R? OR
How to import an Excel or CSV file in R, write all the steps in detail?</td><td>Easy</td></tr>
<tr><td>16.</td><td>Write syntax to check inbuilt dataset in R ;also how to view a particular dataset</td><td>Easy</td></tr>
<tr><td>17.</td><td>Tell the use of any two of following commands from base R package
a. str(dataset name) b. summary(dataset name)
c. View(dataset name)</td><td>Easy</td></tr>
<tr><td>18.</td><td>data.frame() is important command in base R package. Explain this command with syntax and one example</td><td>Easy</td></tr>
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<tr><td>19.</td><td>Create vectors as “Prn No.”, “Name”, “Bank Account”, “Fees paid”, “Mobile No.” a data frame in R for the following table
<table>
<tr><th>Prn No.</th><th>Name</th><th>Bank Account</th><th>Fees paid</th><th>Mobile No.</th></tr>
<tr><td>123B1B01</td><td>Asha</td><td>IOB2205</td><td>147503</td><td>98000046</td></tr>
<tr><td>123B1C55</td><td>Kiran</td><td>IDBI3001</td><td>22572</td><td>54002456</td></tr>
<tr><td>123B1D12</td><td>Rajesh</td><td>SBI4451</td><td>3759</td><td>78940069</td></tr>
<tr><td>123B1D87</td><td>Salman</td><td>AXIS002</td><td>152007</td><td>55987800</td></tr>
<tr><td>123B1L06</td><td>Juhi</td><td>IOB2503</td><td>3759</td><td>22564200</td></tr>
<tr><td>123B1P65</td><td>Ragini</td><td>ICICI123</td><td>154053</td><td>23569800</td></tr>
</table></td><td>Medium</td></tr>
</table>

	Write syntax to create a data frame containing 4 column that includes Roll number 1-10 , Marks of 3 subjects Math, English, Hindi & Marathi (* assume marks)	Medium
21.	Write the script/commands for the following • How to access inbuilt dataset in R? • Select one of the dataset 'storms' and View it • Which commands can be used to find structure of the dataset 'storms' and also write the command to find the statistical summary of the same dataset. Write the command to get first few observations of 'storms' data set	Difficult
22.	Create a script (syntax) for the following steps; I. Import a csv file named 'test.genetics' saved in D drive in folder SDAUR and check structure of it. II. Check data type of all variables. III. Change Maternal.gene type from character to factor and also check is it converted to factor IV. Check first 10 entries of the data. V. Download the file to desktop.	Difficult
23.	Write the step wise process of primary data collection for any one of the following topic a. Effect of Social media on young generation b. Health habits of working women	Difficult